

BPHO ROUND 0 · QUESTION PAPER

Drill bank section 1

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.***Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

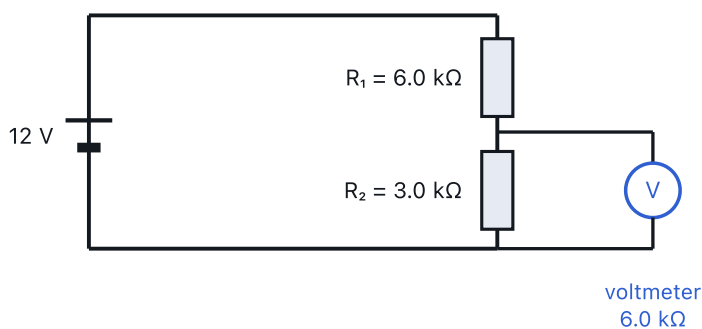
Q1 · Atwood machine: the tension in a string that is not in equilibrium **C**

Two blocks of mass m and $2m$ are joined by a light inextensible string that passes over a smooth fixed pulley, and hang freely. What is the tension in the string while the blocks are moving?

- A** mg **B** $4mg/3$
C $3mg/2$ **D** $2mg/3$
E $8mg/3$

Q2 · A potential divider measured with a voltmeter that is not ideal **H**

A 12 V supply of negligible internal resistance is connected in series with a $6.0 \text{ k}\Omega$ resistor and a $3.0 \text{ k}\Omega$ resistor. A voltmeter of resistance $6.0 \text{ k}\Omega$ is connected across the $3.0 \text{ k}\Omega$ resistor.



What does the voltmeter read?

- A** 1.5 V **B** 2.0 V
C 3.0 V **D** 4.0 V
E 6.0 V

Q3 · Dimensional analysis: the terminal speed of a raindrop against its radius **A**

A spherical raindrop falls through still air and reaches a terminal speed v . The drag force on the drop depends only on the drop's radius r , the density of the air ρ and v itself. The drop is much denser than the air, and the air is much less dense than water (density ρ_w).

How does v depend on r ?

A $v \propto r$

B $v \propto \sqrt{r}$

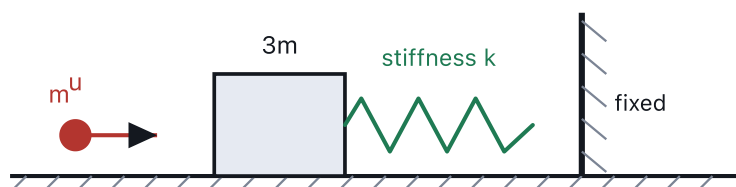
C $v \propto r^2$

D $v \propto 1/\sqrt{r}$

E v does not depend on r

Q4 · An inelastic collision followed by a spring: two conservation laws in a row **C**

A bullet of mass m travelling horizontally at speed u strikes and embeds itself in a block of mass $3m$ resting on a frictionless horizontal surface. The block is attached to a fixed spring of stiffness k .



What is the maximum compression of the spring?

A $u\sqrt{(m/k)}$

B $(u/4)\sqrt{(m/k)}$

C $(u/4)\sqrt{(2m/k)}$

D $2u\sqrt{(m/k)}$

E $(u/2)\sqrt{(m/k)}$

Q5 · Maximum power in a variable load driven by a cell of internal resistance **H**

A cell of emf E and internal resistance r is connected to a variable resistor of resistance R . What is the largest power that can be dissipated in the variable resistor?

A E^2/r

B $E^2/(2r)$

C $E^2/(8r)$

D $E^2/(4r)$

E $E^2 r$

Q6 · Estimation: the mass of the air standing on a square metre **A**

The atmospheric pressure at sea level is about 1.0×10^5 Pa, and **g** is about 10 N kg^{-1} .

Which of these is the best estimate of the mass of the air standing on each square metre of the Earth's surface?

A 10^3 kg

B 10^4 kg

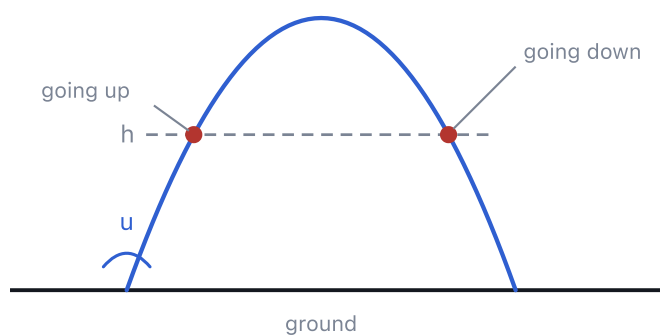
C 10^5 kg

D 10^6 kg

E 10^7 kg

Q7 · A projectile passing the same height on the way up and on the way down **B**

A ball is thrown vertically upward from ground level with speed **u**. It passes a point at height **h** twice: once on the way up and once on the way down.



What is the time interval between the two passages?

A $\sqrt{(u^2 - 2gh)}/g$

B u/g

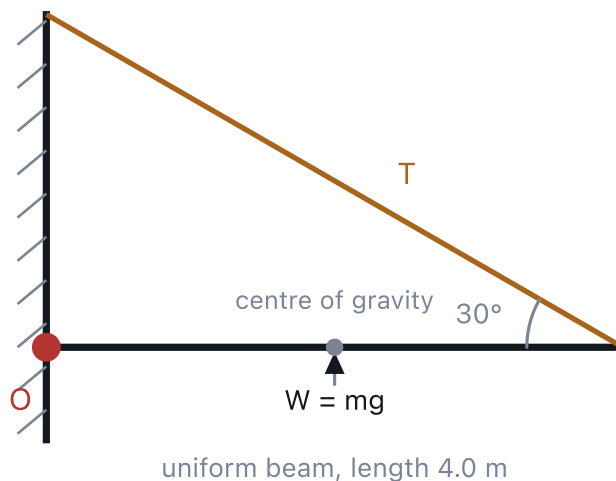
C $2\sqrt{(2gh)}/g$

D $2u/g$

E $2\sqrt{(u^2 - 2gh)}/g$

Q8 · A horizontal beam held by an angled stay: moments about the hinge C

A uniform horizontal beam of mass 20 kg and length 4.0 m is freely hinged to a vertical wall at one end. A light stay wire is attached to the other end and runs to the wall, making an angle of 30° with the beam.



What is the tension in the stay?

- | | |
|----------------|----------------|
| A 50 N | B 100 N |
| C 200 N | D 115 N |
| E 400 N | |
-

Q9 · The speed at which a car leaves the road on a hump-backed bridge D

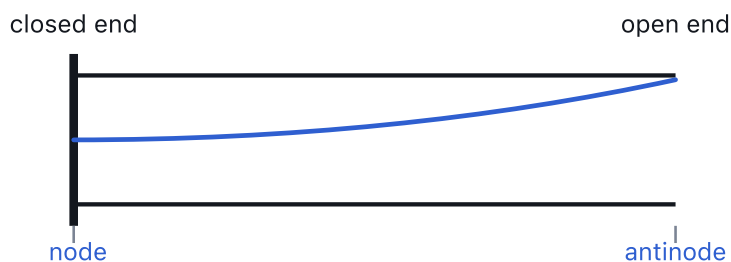
A car crosses a hump-backed bridge whose top may be taken as an arc of radius 20 m. Take $g = 10 \text{ N kg}^{-1}$.

Above what speed does the car leave the road at the top of the bridge?

- | | |
|------------------|------------------|
| A 14 m/s | B 10 m/s |
| C 20 m/s | D 200 m/s |
| E 400 m/s | |
-

Q10 · Three successive resonances of a pipe closed at one end F

A pipe closed at one end is found to resonate at 300 Hz, 500 Hz and 700 Hz, with no resonance in between.



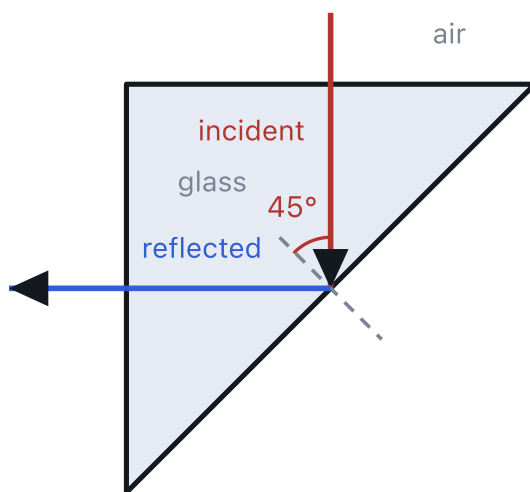
the fundamental: a quarter of a wavelength fits

What is the fundamental frequency of the pipe?

- | | |
|-----------------|-----------------|
| A 50 Hz | B 150 Hz |
| C 200 Hz | D 100 Hz |
| E 300 Hz | |

Q11 · The minimum refractive index for total internal reflection inside a prism G

A ray of light enters a glass prism normally through one of the short faces. The prism is a right-angled isosceles triangle, so the ray meets the long face at 45° .



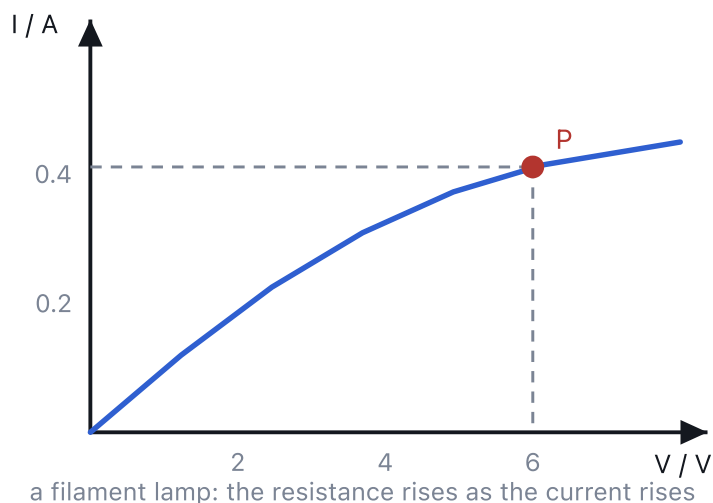
the long face is the hypotenuse

What is the smallest refractive index the glass can have if the ray is to be totally internally reflected at the long face?

- | | |
|---------------------|----------------|
| A 1.2 | B $4/3$ |
| C $\sqrt{2}$ | D 1.5 |
| E $\sqrt{3}$ | |

Q12 · The resistance of a filament lamp at one point on its characteristic H

The graph shows how the current through a filament lamp varies with the potential difference across it. A point on the curve is marked at 6.0 V, 0.40 A.



What is the resistance of the lamp at that point?

- | | |
|-----------------------|-----------------------|
| A 2.4 Ω | B 6.0 Ω |
| C 7.5 Ω | D 15 Ω |
| E 24 Ω | |
-

Q13 · Thermal equilibrium between a hot metal block and cooler water J

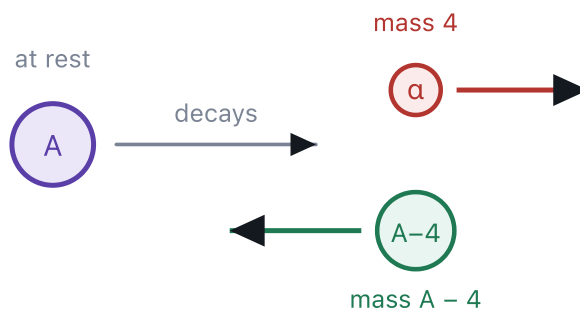
A metal block of mass 0.20 kg and specific heat capacity 500 J kg⁻¹ K⁻¹ is at 200 °C. It is dropped into 0.10 kg of water, of specific heat capacity 4000 J kg⁻¹ K⁻¹, at 20 °C. The container is an ideal insulator.

What is the final temperature of the block and water?

- | | |
|-----------------|-----------------|
| A 20 °C | B 40 °C |
| C 110 °C | D 140 °C |
| E 56 °C | |
-

Q14 · Alpha decay: how the released energy is divided between the two products **K**

A nucleus of mass number **A** is at rest and decays by emitting an alpha particle. The total kinetic energy given to the two products is **E**.



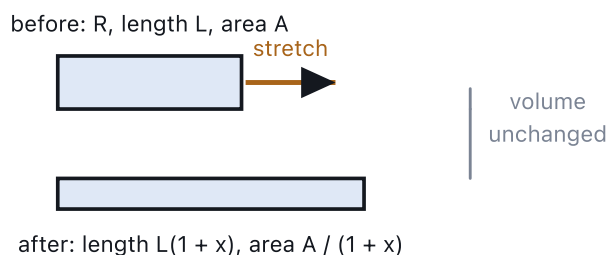
the two products separate with equal and opposite momenta

What is the kinetic energy of the alpha particle?

- A** $E(A - 4)/A$ **B** $4E/A$
C $4E/(A - 4)$ **D** $E/2$
E E

Q15 · Small change: the resistance of a wire stretched at constant volume **A**

A uniform wire of resistance **R** is stretched so that its length increases by a small fraction **x**. The volume of the wire does not change.



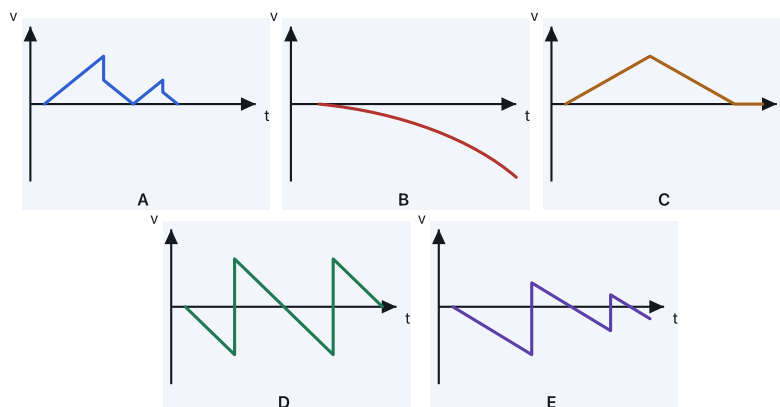
the wire gets longer AND thinner, and both raise the resistance

What is the fractional change in the resistance?

- A** x **B** $2x$
C $x/2$ **D** $4x$
E 0

Q16 · Choosing the velocity-time graph of a bouncing ball **B**

A ball is released from rest and falls to the ground. It bounces, rises to a smaller height than it started from, and bounces again, each bounce being lower than the last. Air resistance is negligible.



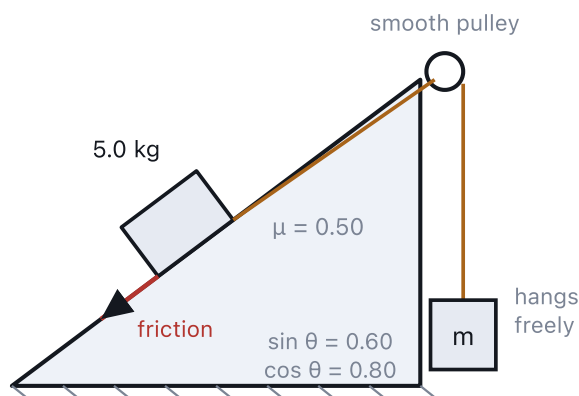
downward is negative

Which graph shows the velocity **v** of the ball as a function of time **t**, taking downward as negative?

- A** — **A** — v falls in a straight line to a negative maximum, then jumps back to the same positive value and the pattern repeats unchanged
- B** — **B** — v falls ever more steeply, becoming negative and growing without limit
- C** — **C** — v rises to a positive maximum, then falls back to zero once and stays there
- D** — **D** — v falls in a straight line, jumps to a positive value, returns to zero, and every bounce reaches the same peak
- E** — **E** — v falls in a straight line, jumps to a smaller positive value, returns to zero, and each bounce is smaller than the one before

Q17 · Limiting friction: the hanging mass that just starts to move a block **C**

A block of mass 5.0 kg rests on a rough plane inclined at an angle whose sine is 0.60 and cosine is 0.80. The coefficient of friction between block and plane is 0.50. A light string runs from the block, up the slope and over a smooth pulley at the top, and hangs vertically with a mass **m** on the end.



Take $g = 10 \text{ N kg}^{-1}$. What is the smallest **m** for which the block just starts to move up the slope?

- | | |
|-----------------|-----------------|
| A 2.0 kg | B 3.0 kg |
| C 4.0 kg | D 5.0 kg |
| E 7.0 kg | |
-

Q18 · Spinning a space station so the rim feels like the Earth's surface **D**

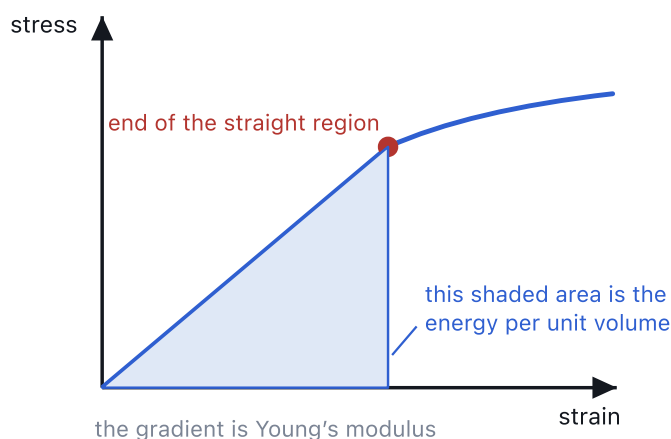
A cylindrical space station of radius 50 m spins about its axis. Take $g = 10 \text{ N kg}^{-1}$.

What period of rotation would make the apparent gravity at the rim equal to g ?

- | | |
|----------------|----------------|
| A 1.4 s | B 2.8 s |
| C 14 s | D 20 s |
| E 31 s | |
-

Q19 · Strain energy per unit volume as the area under a stress-strain graph **E**

The graph shows the stress-strain curve for a metal. The straight-line region ends at a stress of 3.0×10^8 Pa and a strain of 1.5×10^{-3} .

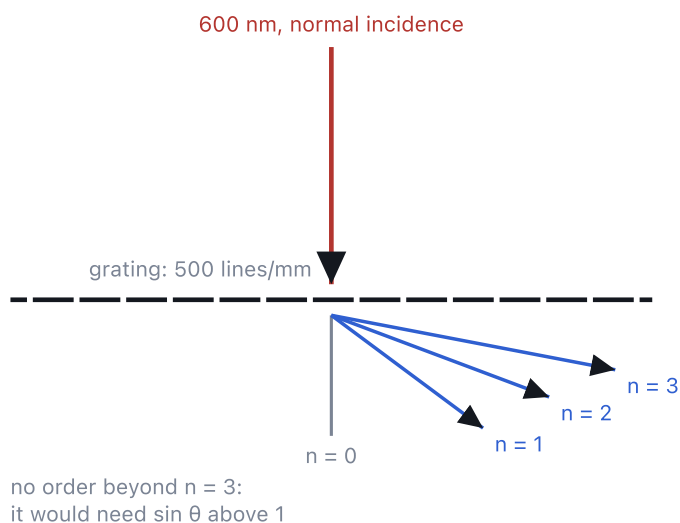


What is the strain energy stored per unit volume at the end of the straight-line region?

- A** $1.1 \times 10^5 \text{ J m}^{-3}$ **B** $2.3 \times 10^5 \text{ J m}^{-3}$
C $4.5 \times 10^5 \text{ J m}^{-3}$ **D** $2.0 \times 10^{11} \text{ J m}^{-3}$
E $4.5 \times 10^2 \text{ J m}^{-3}$

Q20 · The highest diffracted order a grating can produce **F**

Monochromatic light of wavelength 600 nm falls normally on a diffraction grating ruled with 500 lines per millimetre.

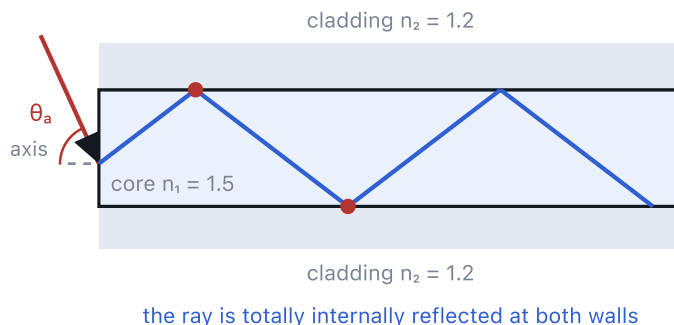


What is the highest order of diffracted beam that can be observed?

- A** 3 **B** 4
C 6 **D** 7
E 12

Q21 · The acceptance angle of a step-index optical fibre **G**

A step-index optical fibre has a core of refractive index 1.5 and a cladding of refractive index 1.2. Light enters the flat end face from air.



What is the sine of the largest angle of incidence at the end face for which light is still guided along the fibre?

- | | |
|---------------|--------------|
| A 0.45 | B 0.9 |
| C 0.8 | D 1.0 |
| E 1.2 | |

Q22 · How long a kettle takes: electrical power meeting thermal capacity **H**

An electric kettle is rated at 2.0 kW. It contains 1.5 kg of water at 20 °C. The specific heat capacity of water is 4200 J kg⁻¹ K⁻¹, and all the electrical energy goes into the water.

How long does the water take to reach 100 °C?

- | | |
|----------------|----------------|
| A 25 s | B 126 s |
| C 315 s | D 504 s |
| E 252 s | |

Q23 · A charged capacitor connected to an uncharged one: where the charge goes **I**

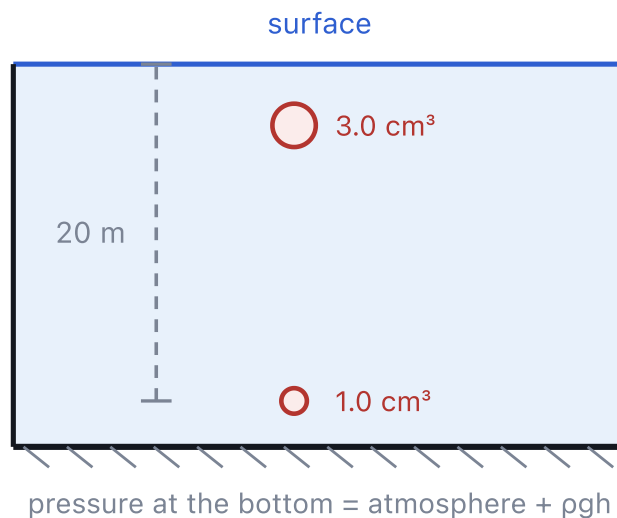
A 2.0 μF capacitor is charged to 100 V and then disconnected from the supply. It is connected in parallel with an uncharged 3.0 μF capacitor.

What is the potential difference across the pair once the charge has settled?

- | | |
|----------------|---------------|
| A 20 V | B 50 V |
| C 40 V | D 60 V |
| E 100 V | |

Q24 · A bubble rising in a lake: pressure at depth against volume **J**

A bubble of volume 1.0 cm^3 is released from the bottom of a lake 20 m deep. The atmospheric pressure at the surface is $1.0 \times 10^5 \text{ Pa}$, the density of water is 1000 kg m^{-3} , and $g = 10 \text{ N kg}^{-1}$. The temperature is the same at the bottom and at the surface.



What is the volume of the bubble when it reaches the surface?

- | | |
|-----------------------------|-----------------------------|
| A 3.0 cm^3 | B 2.0 cm^3 |
| C 4.0 cm^3 | D 1.5 cm^3 |
| E 1.0 cm^3 | |

Q25 · What survives one and a half half-lives **K**

A radioactive source has half-life **T**.

What fraction of the original nuclei remain undecayed after a time **$1.5T$** ?

- | | |
|----------------|--------------------------|
| A $1/2$ | B $1/4$ |
| C $3/4$ | D $1/(2\sqrt{2})$ |
| E $1/8$ | |